

Korea Foreign Ownership - Research Record

Working record of the foreign-ownership flow research. Every number here was computed by the pipeline in this repo from the exchange's end-of-day files; nothing is hand-entered. Negative results are kept -- they are the answer, not the absence of one.

store	fields used	how the field is used
foreign_ownership	foreign_pct, foreign_shares	the signal: level and daily change of foreign holding per issuer
foreign_ownership	shares_listed, foreign_limit_shares	ownership ratio denominators, statutory-limit usage
foreign_ownership	market	KOSPI / KOSDAQ split
market	close, value_traded	forward returns, ADV liquidity filter, KRW valuation of flows
market	market_cap	value-weighted market aggregates

Coverage: ~3,600 issuers x ~4,100 trading sessions, collected daily on a 19:00 KST schedule and stored as immutable per-day parquet.

Step 1 - Build the history

Question: can a clean per-issuer daily history be built at all?

Yes. The exchange publishes an end-of-day foreign-holding file per issuer; the collector fetches missing days idempotently, so a skipped run never becomes a gap. This dataset is the project's primary deliverable and every later step consumes it.

item	value
issuers	~3,600
sessions	~4,100
refresh	daily 19:00 KST, self-publishing

```
# Step 1 - collection pipeline (01_backfill.py, 05_daily_update.py)
```

Step 2 - Persistence (the core hypothesis)

Question: is foreign flow autocorrelated, as the brief guessed?

Yes, strongly. Cross-sectional rank autocorrelation of daily flow at lag 1 is +0.088 with t +64.7, and it survives a tie-artifact control (98% of the effect retained). The premise of the whole idea is true.

lag	rank AC
1	+0.088 (t +64.7)
tie-artifact control	98% retained

```
# Step 2 - persistence (04_validate.py)
```

Step 3 - Does it predict returns?

Question: does today's foreign flow predict tomorrow's cross-section?

Only nominally. The 1-day rank IC is +0.0039 (t +3.9) -- real but tiny -- and by 20 days the sign flips (-0.0039, Newey-West t -4.2). Whatever is being anticipated at one day is given back within a month.

horizon	rank IC	t
1d	+0.0039	+3.9
20d	-0.0039	-4.2 (NW)

```
# Step 3 - horizon IC (04_validate.py)
```

Step 4 - Where does the effect live?

Question: is the 1-day effect in names that can actually be traded?

No. Split by liquidity, the effect concentrates in the illiquid tail. In the liquid top third of the market -- the only place size could be deployed -- the IC is statistically zero.

universe	1d IC	t
all names	+0.0039	+3.9
liquid top third	+0.0007	+0.7

```
# Step 4 - liquidity split (12_combination.py)
```

Step 5 - Which t-statistics can be trusted?

Question: daily ICs are autocorrelated (the signal itself persists) -- which standard error is honest?

A two-arm null calibration: feed each estimator data that is truly null under (a) an MA(h) process and (b) white noise, and check its rejection rate. Naive SEs over-reject under (a); a non-overlapping estimator under-rejects to 0% under (b). Newey-West is the only one calibrated under both, so every headline t in this record is NW. (The non-overlapping estimator was my first choice; the calibration showed it was the worst of the three, and the verdict was switched.)

```
# Step 5 - SE calibration (09_overlap_correction.py, 09b_se_diagnostic.py)
```

Step 6 - Level signals

Question: does the LEVEL of foreign ownership (not its change) carry alpha?

It looks like it until autocorrelation is priced in. The 60d level IC has a naive t of +35.9, but the daily IC series has $AC(1) = 0.974$: Newey-West cuts the t to +5.95 and the effective sample is 43 observations out of 1,567. That is a handful of regimes, not an edge -- classified as regime exposure under the rule set before the test was run.

signal	naive t	NW t	n_eff
level 60d	+35.9	+5.95	43 / 1,567

```
# Step 6 - level signals (11_level_se.py)
```

Step 7 - Costs

Question: does anything survive real Korean trading costs?

No. The 1-day signal needs ~320 rebalances a year at 63.5% turnover each; at 30bp round trip plus the 0.20% securities transaction tax the long-short book loses -89.7% a year. The signal decays faster than it can be traded.

item	value
turnover per rebalance	63.5%
rebalances / yr (1d)	~320
net at 30bp + tax	-89.7% / yr

```
# Step 7 - cost model (04_validate.py section 2c)
```

Step 8 - The comparison nobody asked for

Question: if persistence is the reason to follow a flow, is FOREIGN flow the most persistent one?

No -- domestic institutions are. Measured on the same basis, institutional flow autocorrelation is ~2.3x the foreign figure at every lag out to 20. The brief's mechanism ("foreign investors digest information slowly") is refuted in its comparative form; the persistence itself is real but foreign flow is not where it is strongest.

series	lag-1 AC	t
domestic institutions	+0.298	+235.9
foreign	+0.130	+95.2

```
# Step 8 - investor comparison (04_validate.py section 5)
```

Step 9 - Index-rebalance denoising

Question: can MSCI/FTSE rebalance noise be filtered out, as planned?

Not with a calendar. Measured flow on 136 known effective dates is 1.02x a normal day -- indistinguishable. Either the flow spreads across surrounding days or the store's holding series absorbs it differently; either way there is no event spike to filter, so this brief item is blocked, not skipped.

test	result
effective-date flow multiple	1.02x (136 dates)

```
# Step 9 - rebalance calendar test (krxflow/rebalance.py)
```

Step 10 - Orthogonality and combination

Question: the brief's real intent -- is this useful ALONGSIDE other features?

Half-answered. The signal survives residualisation against size, liquidity, reversal, momentum and volatility, so its information is not already in the standard set -- the property that matters for combination. But the factor composite built here as a stand-in baseline is itself unprofitable (IC -0.078), so a marginal-contribution number on top of it would be meaningless. The honest statement: combination cannot be judged without the real production feature stack. That test is the natural next step and needs access, not more data.

```
# Step 10 - orthogonality (12_combination.py)
```

Step 11 - The live tracker

Question: what would riding the flow actually have earned? (recomputed on every refresh)

Buy the top decile of 20-session accumulation, hold h sessions, versus the equal-weighted universe, 0.50% per round trip. Gross is statistically zero at every horizon and at best a quarter of the cost hurdle; net is negative everywhere. This table lives in docs/monitor.html and re-runs itself, so the verdict stays current instead of being a one-time backtest.

hold	trades	gross bp	net bp	t (NW)	hit	net /yr
1d	619	+0.7	-49.3	0.30	51%	-123%
5d	123	+4.2	-45.8	0.34	55%	-23%
20d	30	+12.8	-37.2	0.28	57%	-4.6%

Step 11 - forward tracker (tracker.py, monitor section)

Step 12 - Open data puzzle

Question: does the holdings-implied flow match the exchange's own reported net buying?

Only at +0.10 correlation across ~4,100 days x ~3,600 issuers. Candidate explanations: settlement vs trade date, securities lending, custody transfers, DR conversions, in-kind index creation. It does not change any verdict above, but it changes what the series measures. Next data step: collect the official investor-type net-buying series (free) alongside.

Step 12 - holdings vs reported flow cross-check

Every design decision, side by side

decision	alternatives tested	chosen	why
standard error	naive / non-overlapping / Newey-West	Newey-West	only one calibrated under both nulls
signal form	daily change / level / 60-120d change	none deployable	level = regime exposure; changes die at cost
universe	all / liquid top third	liquid third for any trading claim	effect elsewhere is untradable
cost model	zero / 30bp / 30bp + 0.20% tax	30bp + tax	Korean sell-side tax is unavoidable
forward returns	raw / split-guarded	split-guarded	unadjusted prices fake -90% days

Bottom line

- **Dataset (deliverable): done and self-maintaining.** Daily per-issuer foreign ownership, published to the repo on a daily schedule.
- **Hypothesis: premise confirmed** (persistence is real, t +64.7), but the comparative story is inverted -- institutions are 2.3x more persistent.
- **Standalone trading: refuted three ways** (untradable universe, 20d sign flip, costs), and a live tracker keeps re-checking it.
- **Combination with production features: the remaining question**, and the one the dataset was built for. Needs feature-stack access, not more data.